Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

The dataset for this analysis consists of the monthly taxi trip records for the year of 2023. So, each month's data was stored in a separate Parquet file, these files were loaded using the Pandas library. A loop was used to read all 12 files systematically, ensuring efficient handling of the provided dataset.

1.1.1. Sample the data and combine the files

As there was a huge amount of data given, a 5% of random sample was taken from each monthly file to improve the computational efficiency while still maintaining the representation of the given data.

All the sampled datasets were then combined together into a single data frame using concatenation. This consolidated dataset was used for the future analysis.

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

As the dataset columns were reviewed to ensure that the correct data types and their consistency. Date and time columns were converted into datetime format to enable the time-based analysis.

2.1.2. Combine the two airport_fee columns

The given dataset contained duplicate columns for airport fees with different naming conventions (airport_fee and Airport_fee). These were combined into a single column to maintain the consistency and avoid the redundancy.

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

Missing values were identified across multiple columns such as the passenger count, RatecodeID, and the surcharge-related fields. The proportion of the missing values was analyzed to understand which appropriate method to be used.

2.2.2. Handling missing values in passenger_count

Missing or zero values in the passenger_count column were taken care of, by either removing invalid entries or retaining only meaningful observations where passenger counts were greater than zero.

2.2.3. Handle missing values in RatecodeID

Here the missing values in the RatecodeID column were handled by either imputing with the most frequent category or leaving them as unknown was appropriate, as this variable had limited impact on the main analysis.

2.2.4. Impute NaN in congestion_surcharge

Missing values in the congestion_surcharge column were imputed using appropriate default values or the statistical measures, depending on the distribution of the data.

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

Several unrealistic values were seen in the dataset, such as extremely high trip demands, negative fares, and abnormal tip values. These were treated as the outliers and removed to improve the quality of data.

Filters were applied to retain only correct observations, like:

- Positive fare and trip distance,
- Reasonable trip speeds (less than 100 mph),
- Logical fare per mile value, and
- Tip percentage within realistic limits.

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

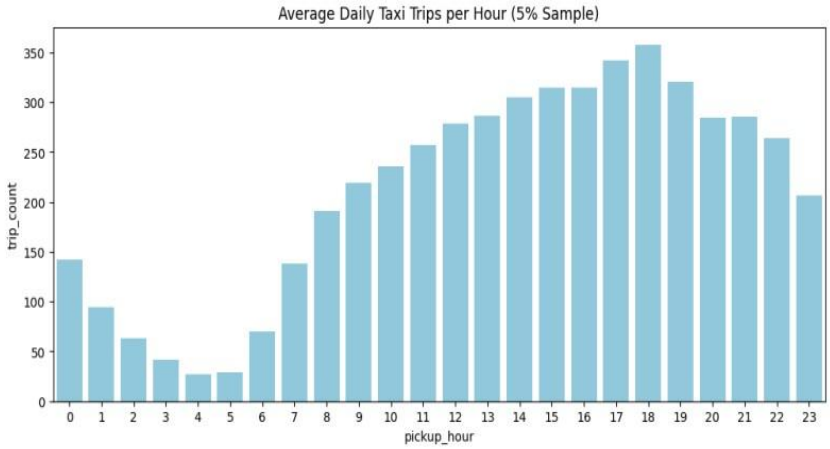
3.1.1. Classify variables into categorical and numerical

The dataset variables were classified into two main categories:

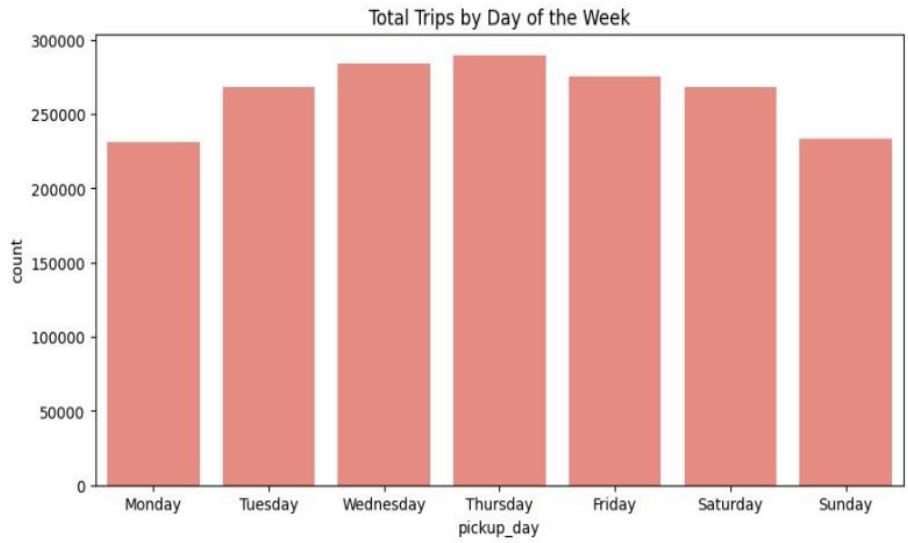
- Categorical variables: payment type, pickup day, pickup month, passenger count.
- Numerical variables: fare amount, trip distance, tip amount, trip duration.

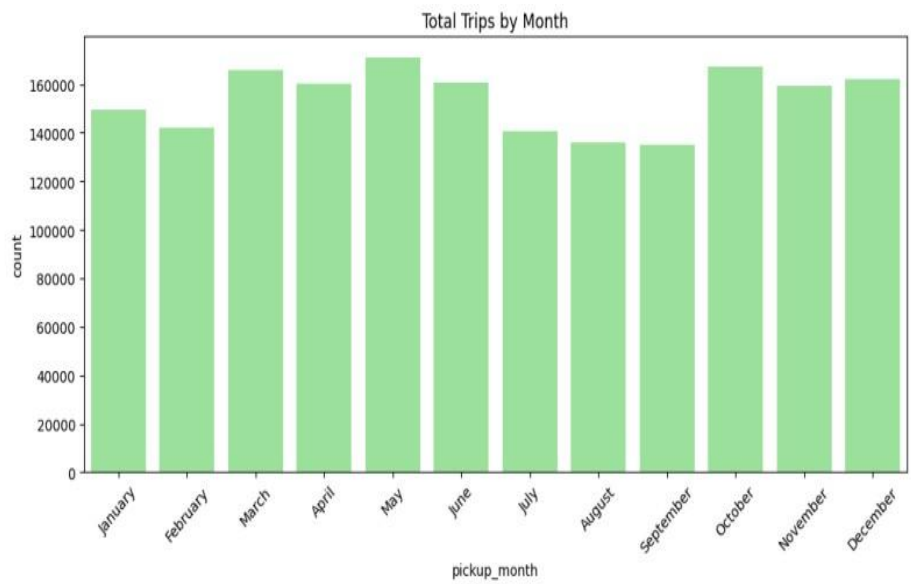
3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months

The distribution of the taxi trips was analyzed across different time dimensions such as hours, days, and months. To understand how taxi



demand varies over time, the distribution of trips was analysed across different hours of the day, days of the week, and months of the year.





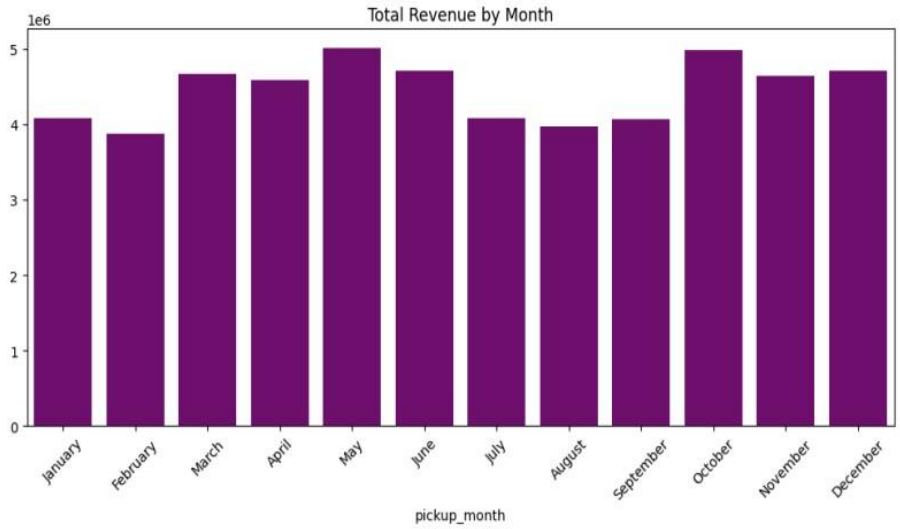
It was noticed that the taxi demands vary significantly throughout the day, with peak activity during typical commuting hours. Weekdays showed more consistent demand, while weekends displayed slightly different patterns. Monthly trends indicated seasonal fluctuations in using the taxi.

3.1.3. Filter out the zero/negative values in fares, distance and tips

Trips with zero or negative values in fare, distance, or tips were removed to ensure the accuracy in the analysis and avoid unexpected results.

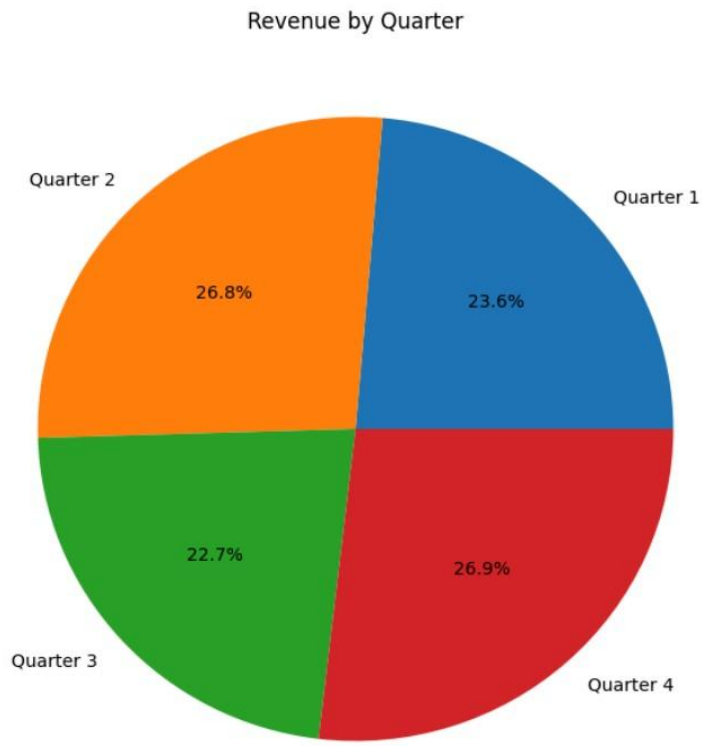
3.1.4. Analyse the monthly revenue trends

The Monthly revenue trends were analyzed using the total trip amount. The results indicate variations in revenue across different months, possibly due to seasonal demand patterns and external factors such as holidays and weather conditions.



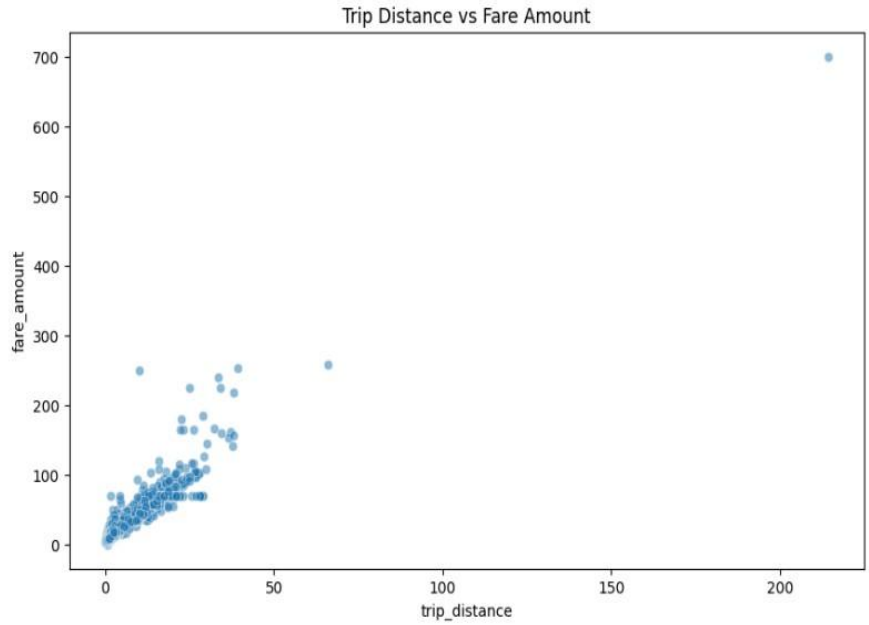
3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

The revenue was also analyzed on a quarterly basis to understand the big picture. The contribution of each quarter to the total yearly revenue was calculated and visualized, highlighting periods of the higher financial performance.



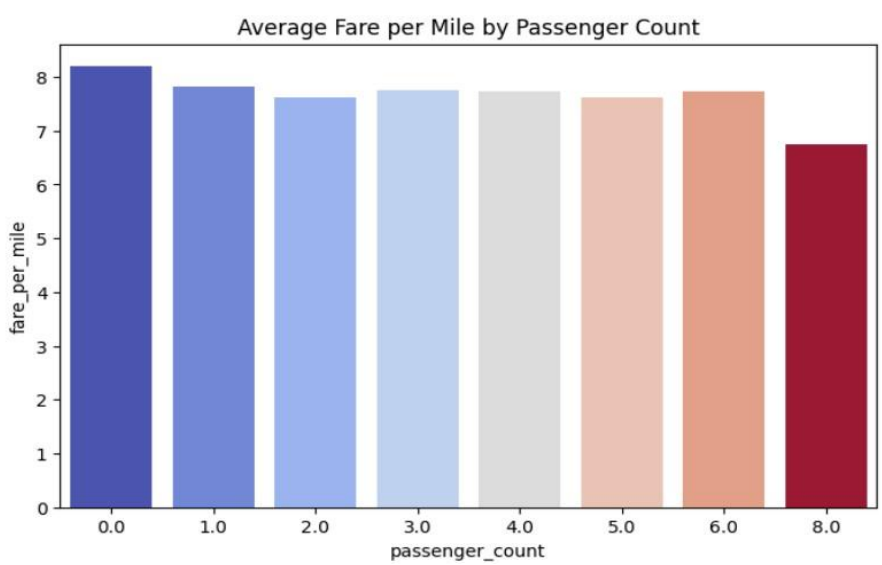
3.1.6. Analyze and visualise the relationship between distance and fare amount

A scatter plot was used to analyze the relationship between trip distance and fare amount. It is noticed that the fare generally increases with distance, indicating a positive correlation between the two variables. However, the relationship is not perfectly linear, as fare calculation includes base charges, surcharges, and time-based components. Some variability in fares for similar distances tells the influence of traffic conditions and additional fees.



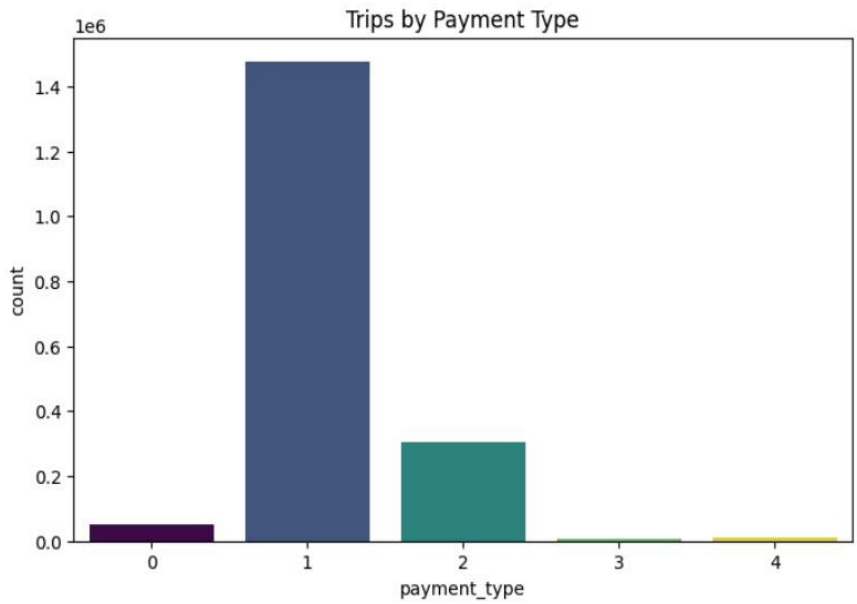
3.1.7. Analyse the relationship between fare/tips and trips/passengers

Here the relationship between fare, tips, and passenger count was analyzed to understand customer behavior. It was understood that trips with a higher number of passengers tend to have slightly more total fares. However, the tip amounts did not increase proportionally with passenger count. This indicates that tipping behavior is influenced more by service quality and payment method rather than the number of passengers.



3.1.8. Analyse the distribution of different payment types

The distribution of payment types showed that certain payment methods such as card payments were most commonly used compared to cash. Here it can be understood that passengers growing preference for the cashless transactions. Additionally, tip data is more recorded continuously for card payments, making them more reliable.



3.1.9. Load the taxi zones shapefile and display it

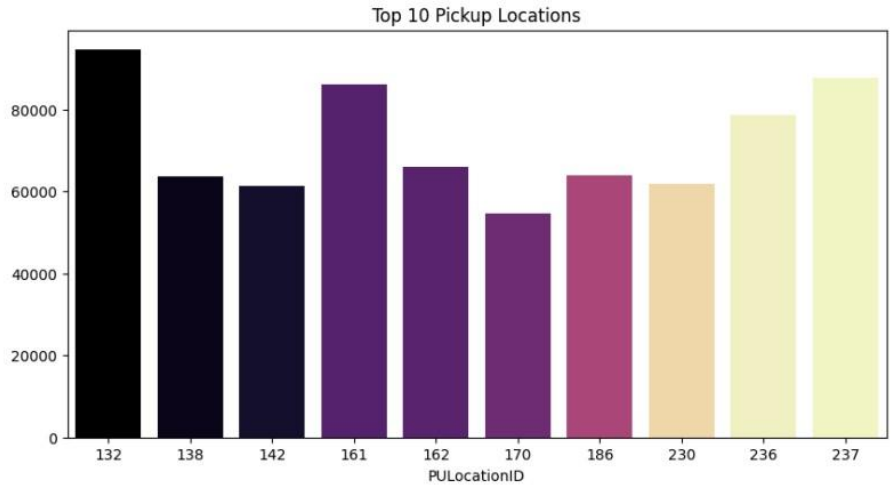
The taxi zones shapefile was loaded to obtain geographical boundaries of different zones in New York city. This data helps us to spatially analyze the taxi activity around different regions.

3.1.10. Merge the zone data with trips data

The taxi trip dataset was merged together with the zone shapefile using location IDs. By doing this it helps to map the trip data to specific geographical zones for further spatial analysis.

3.1.11. Find the number of trips for each zone/location ID

The total number of trips for each zone was calculated using pickup location IDs. This helped to identify the high- demand areas with the most taxi activity.

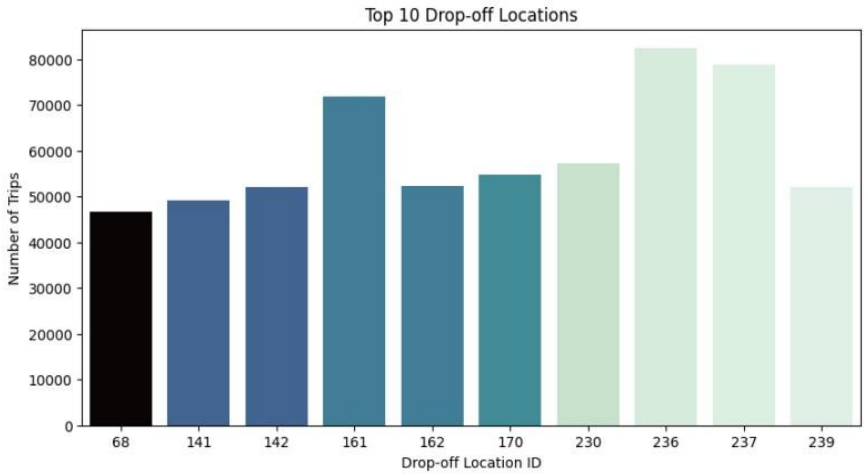


3.1.12. Add the number of trips for each zone to the zones data frame

The calculated trip counts were added to the zone's dataset, enabling to directly compare the activity of the taxi around different regions.

3.1.13. Plot a map of the zones showing number of trips

A map visualization was created to display the number of trips across different zones. Zones with more trip counts were clearly distinguishable, highlighting areas with higher demand.



3.1.14. Conclude with results

The EDA (Exploratory data analysis) showed important patterns in taxi usage across time, location, and customer behavior. The most important observation includes peak travel hours, high- demand zones, and the dominance of digital payment methods. These observations form the basis for the future detailed analysis and strategy development.

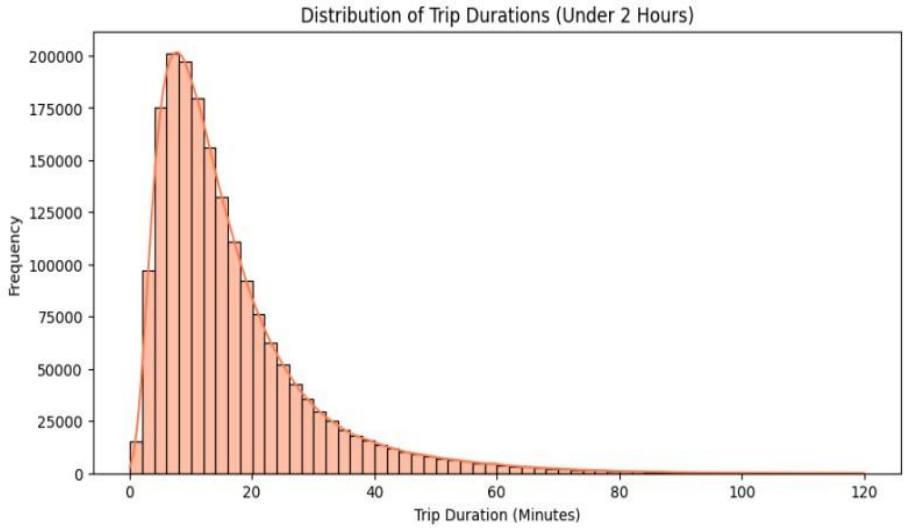
3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

Here the routes with less average speeds were identified by understanding the trip distance and the duration. These slow routes are likely affected by the traffic congestion or the road conditions. Identifying such routes can help us to better optimize routing and reduce further delays.

3.2.2. Calculate the hourly number of trips and identify the busy hours

The number of trips was analyzed on an hourly basis. Peak demand was observed during morning and in the evening hours, corresponding to a typical work commute time.

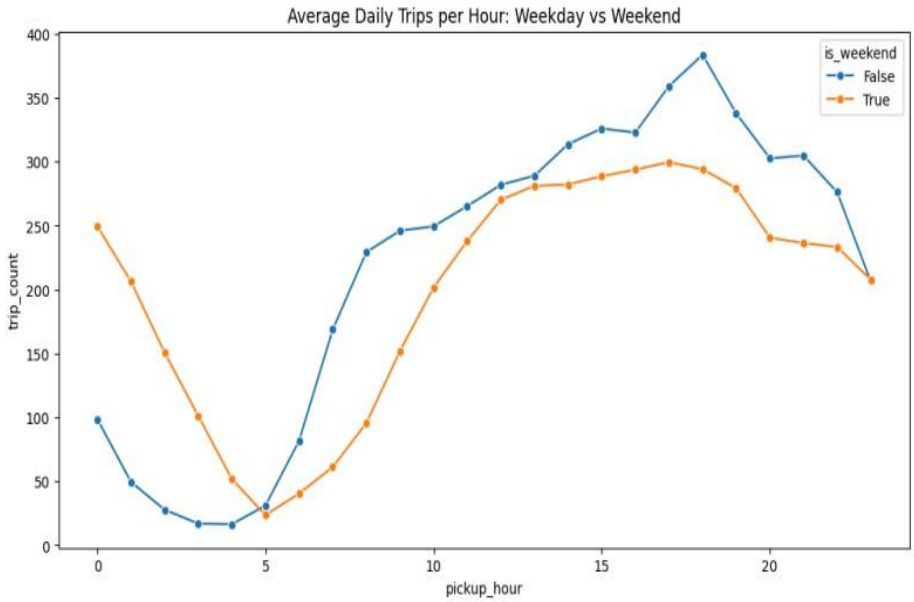


3.2.3. Scale up the number of trips from above to find the actual number of trips

Since only a 5% of the sample of the data was used, the actual number of trips was estimated by scaling the observed values. This gives a more realistic estimation of the total taxi demand across the city.

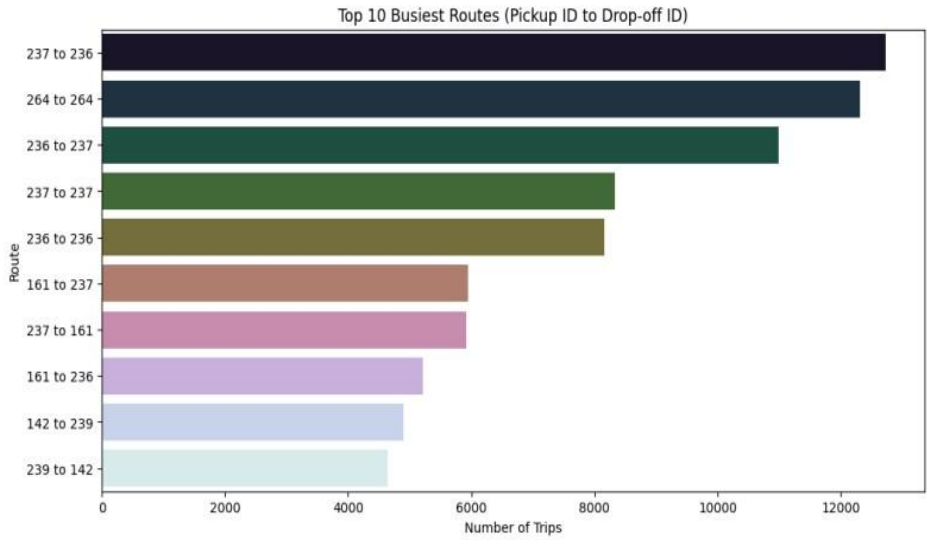
3.2.4. Compare hourly traffic on weekdays and weekends

A comparison of the hourly traffic patterns between weekdays and weekends showed differences in their travel behavior. Weekdays have more demand during the working hours, while weekends showed increased activity during late hours and leisure times.



3.2.5. Identify the top 10 zones with high hourly pickups and drops

The top 10 zones with the highest pickup and drop- off activity were identified. These zones represent key areas of high demand and should be given more priority for the availability of the taxi.



3.2.6. Find the ratio of pickups and drop-offs in each zone

The ratio of pickups to drop-offs in each zone were analyzed to understand traffic flow. Zones with higher pickups than drop-offs may indicate residential or the commercial hubs, while the opposite suggests destination areas.

3.2.7. Identify the top zones with high traffic during night hours

Certain zones showed more activity during the night time hours. These areas likely to represent entertainment hubs, airports, or late-night service areas.

3.2.8. Find the revenue share for nighttime and daytime hours

The revenue generated during the daytime and night time was compared. It was observed that while daytime as a larger share due to the volume, night time trips may have higher fares due to surcharges.

3.2.9. For the different passenger counts, find the average fare per mile per passenger

The average fare per mile was analyzed across different passenger counts. The results indicate that the fare per mile remains relatively stable, suggesting the price consistency regardless of the passenger

count.

3.2.10. Find the average fare per mile by hours of the day and by days of the week

Fare per mile was understood around different hours and days. Slight variations were observed, may be due to the demand fluctuations and traffic conditions.

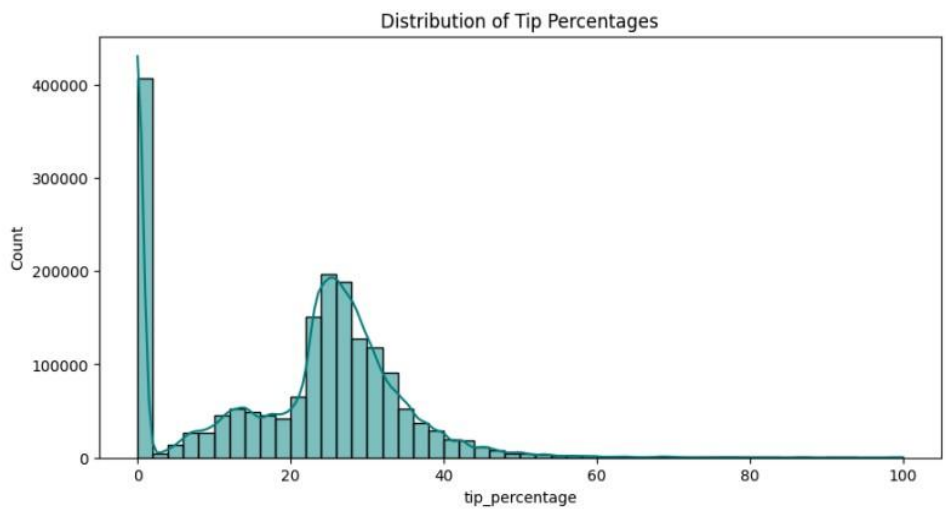
3.2.11. Analyse the average fare per mile for the different vendors

The average fare per mile was compared across different vendors. Small differences were noticed, indicating a similar pricing strategy among vendors.

3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion

The fare rates were compared across vendors for different distance ranges. This analysis helps identify whether certain vendors charge differently for short or long trips.

3.2.13. Analyze the tip percentage

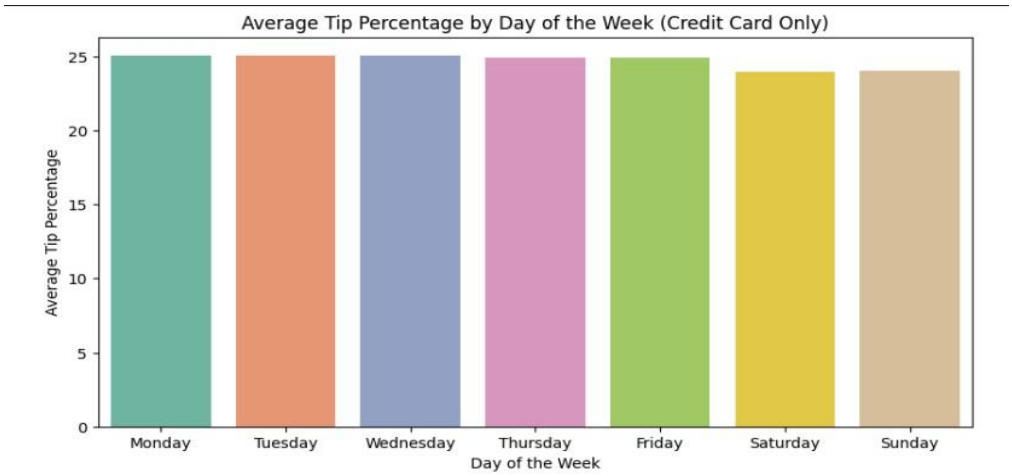


The distribution of tip percentages shows that the most values are concentrated within a moderate range, with very few high values. This

indicates that the passengers generally follow a consistent tipping pattern rather than giving very low or very high tips.

The presence of a clear concentration suggests that tipping behaviour is relatively stable across most of the trips. Extremely high or low tips are rare and may be due to special circumstances.

Since tipping behaviour is consistent, improving service quality and encouraging digital payments can help maintain or slightly increase tip amounts for drivers.



The average tip percentage varies slightly across the different days of the week. This analysis considers only the credit card transactions, as tip data is more accurately recorded for such payments.

Some days show marginally higher tipping behaviour, which may be influenced by the factors such as trip purpose, customer mood, or travel patterns during those days.

Drivers can be encouraged to work more during the days with higher average tips. Additionally, promoting card-based payments can help ensure better tracking and consistency of tip data.

3.2.14. Analyse the trends in passenger count

The trends in passenger count indicated that most trips only involved a single passenger. Higher passenger counts are less very often.

3.2.15. Analyse the variation of passenger counts across zones

The variation in passenger counts across various zones were analyzed. Some zones showed higher group travel, possibly indicating tourist or commercial areas.

3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

The application of extra charges such as surcharges was analyzed across different zones and time periods. These charges were more frequent during the peak hours and in the high- demand zones.

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

From this analysis, it is clear that the taxi demand is highly time- dependent, with more peak activity noticed during morning and evening hours. At the same time, certain routes were identified to have significantly fewer average speeds, indicating congestion or inefficiencies.

To optimize routing, taxi operations should prioritize faster and more efficient routes during peak hours instead of relying on commonly used but congested paths. Integrating real-time traffic data into routing decisions can help drivers avoid delays and complete more trips within the same time limit.

In terms of dispatching, taxis should be dynamically allocated based on the demand patterns observed across different hours of the day. For example, increasing fleet availability during the peak hours and reducing idle taxis during low-demand periods can be able to improve overall utilization of the taxi.

This approach can lead to reduced waiting times for passengers and increased trip frequency for drivers, thereby improving both service efficiency and revenue generation.

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

The spatial analysis revealed that the demand for the taxi is not evenly distributed around all the zones. A small number of zones consistently recorded high pickup and drop-off activity, making them the most important demand hotspots.

To make efficient use of the resources, taxis should be strategically positioned in these high-demand zones, especially during the peak hours. For example, business districts are likely to have higher demand during weekdays, while entertainment or leisure zones show increased activity during the evenings and weekends.

Additionally, zones with high drop-off but relatively lower pickup activity can be monitored to repositioning of the taxis proactively, ensuring that the vehicles are not left idle in low-demand areas.

By aligning the availability of the taxis with both spatial and temporal demand patterns, operators can minimize idle time, reduce unnecessary fuel consumption, and improve overall service coverage across the city.

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

The analysis of fare patterns, tip behavior, and trip characteristics provides useful understanding for pricing strategy. It was observed that while fare generally increases with distance, there are variations influenced by time of the day, demand levels, and trip conditions.

To maximize the revenue while remaining competitive, a dynamic pricing approach can be taken into the consideration. During peak demand hours, slightly higher pricing can be applied to reflect increased demand and limited supply. At the same time, maintaining competitive base fares ensures that customers are not discouraged from using taxi services.

Additionally, higher tip percentages observed in card-based transactions indicate the importance of promoting the digital payments. Encouraging such payment methods can indirectly improve driver earnings.

Pricing strategies can also be adjusted based on trip distance tiers, ensuring that both short and long trips remain profitable without overcharging customers.

Overall, a balanced pricing model that considers demand fluctuations, customer behavior, and market competition can help improve revenue while maintaining the satisfaction of the customer.